

Zone Submission – Media Forensics & Deepfake Lab

1. Zone Idea

Concept: A 3-hour hands-on forensic investigation lab where registered participants analyze manipulated media (images, video, audio) using real forensic tools. Participants solve three progressively challenging tasks: detecting a cloned area via Error Level Analysis (ELA), de-blurring a license plate from a video, and extracting a hidden message from an audio spectrogram.

Theme: “You are the digital forensic analyst – uncover the truth hidden in pixels, frames, and sound.”

Why it works:

Structured 3-hour session ensures deep engagement.

Real forensic techniques (ELA, frame extraction, spectrogram analysis).

Immediate reward (flags, prizes) per solved challenge.

Runs in registered batches to manage capacity.

2. Workflow

Before the event

Prepare challenge files, tools, and HTTP server on zone laptop.

Set up registration list (time slots: e.g., 9:00–12:00, 13:00–16:00, etc.).

Print score sheets, challenge summaries, and prize tokens.

During a batch (3 hours)

Time	You Do	Participants Do
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0:00–0:15	Intro, share IP, start HTTP server	Connect to Wi-Fi, download all files via browser
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0:15–0:25	Explain Challenge 1 (ELA) + live demo	Listen, ask questions
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0:25–1:15	Walk around, provide hints	Solve Challenge 1 (find cloned area, submit flag)
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1:15–1:25	Reveal answer, explain Challenge 2 (de-blur)	Check answer, listen
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1:25–2:15	Walk around, provide hints	Solve Challenge 2 (extract frames, sharpen, read plate)
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2:15–2:25	Reveal answer, explain Challenge 3 (spectrogram)	Check answer, listen
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2:25–2:55	Walk around, provide hints	Solve Challenge 3 (extract audio, find hidden word)
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2:55–3:00	Reveal all answers, final Q&A, distribute prizes	Submit flags (verbally or via Google Form)
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After each batch

Clear score sheet, prepare for next batch (15-min buffer).

Restart server if needed, restock prizes.

3. What Things You Need

Hardware

1 laptop (yours) – runs HTTP server, has all files and tools.

1 backup laptop – with same files (in case of failure).

Projector / large screen – for demos and reveals.

Power strip / extension cord – for your laptop and participants' laptops.

Small table and chairs – for 10–15 participants.

Software & Files (pre-loaded on your laptop)

Python (to run `python -m http.server`).

Challenge files (3 folders – one per challenge).

Portable tools (FFmpeg, Audacity Portable, GIMP Portable, StegSolve.jar, ExifTool).

HTTP server – started with one command.

Printed Materials

Score sheets (one per batch – participant name, checkboxes for each challenge).

Challenge summary sheet (one per participant – brief instructions, commands).

QR code linking to flag submission form (Google Form) – optional.

Prizes

Stickers, candy, small tokens – enough for 3 prizes per participant × number of batches × batch size.

4. Budget

Item	Estimated Cost (INR)	Notes
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Prizes (stickers/candy)	₹500 – ₹1,000	100–150 items
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Printed materials (score sheets, summaries)	₹200 – ₹300	A4 paper, black & white
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USB sticks (backup, optional)	₹1,000 – ₹2,000	10–20 sticks
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Total	₹1,700 – ₹3,300	(~\$20–\$40 USD)
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If the event already provides projector, table, and basic stationery, budget can be as low as ₹500–₹1,000 (prizes only).

5. How Many Volunteers Needed

Role	Number	Responsibilities
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Zone Lead (you)	1	Run the 3-hour session, answer technical questions, manage HTTP server, give hints, reveal answers, manage registration check-in.
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Assistant (optional)	1	Help distribute prizes, reset score sheets, assist participants who are stuck, manage crowd during batch transition.
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If batch size ≤ 10 : You can manage alone.

If batch size 10–15: An assistant is highly recommended.

6. How Many Participants Can Experience Your Zone at a Time

Maximum per batch: 15 participants.

Reason: You can effectively help 15 people during the hands-on phases. More than that becomes chaotic.

Total per day: Depends on number of batches (e.g., 3 batches = 45 participants, 4 batches = 60 participants).

Batch schedule example:

Batch	Time	Capacity
1	9:00 – 12:00	15
2	13:00 – 16:00	15
3	16:00 – 19:00	15
Total daily capacity		45 participants

(Adjust based on event duration.)

Additional Notes for Organizers

Registration required – no walk-ins. Participants sign up for a time slot.

Participants bring their own laptops with Wi-Fi capability.

Internet is optional – all analysis can be done offline after downloading files.

The zone is self-contained – does not rely on external servers (except optional Google Form for submissions).